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A Benchmark of Five Machine-Learning Architectures for Daily Trading-Signal Classification on US Tech Equities

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Do deep networks beat simple models?

Daily BUY / HOLD / SELL signals for seven large US tech stocks

What is often missing in the literature

1

Only 1–2 models

Few studies compare deep models against a simple linear baseline.

2

Accuracy or RMSE

These metrics do not tell if a trading strategy makes money.

3

Untested feature filters

Statistical pre-selection is applied without checking that it helps.

Our goal: a fair five-model benchmark, scored as classification and as a trading strategy.

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One signal per day, seven tech stocks

Free daily data · adaptive three-class label

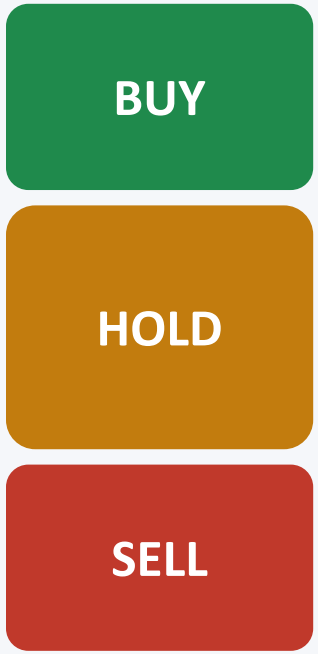
Stocks

AAPL · NVDA · TSLA · AMZN

MSFT · GOOGL · META

- **Yahoo Finance,** free, 2013–2025
- **61 features** in 9 families
- **Market context:** S&P 500 (SPY) and VIX

Label for day t (tomorrow's return vs. last 252 days)



Adapts to volatility · uses only past data · ≈ 30 / 40 / 30 %

Five models, one protocol

Same data, same features, same test year

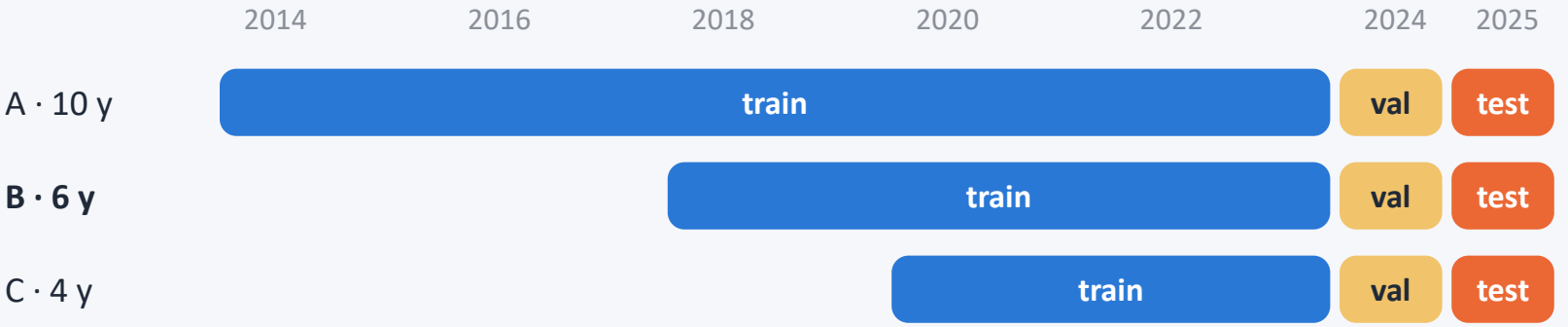
Classical

- Logistic reg.
- XGBoost

Deep learning

- LSTM
- 1-D CNN
- CNN-LSTM

Three splits, same test year



120

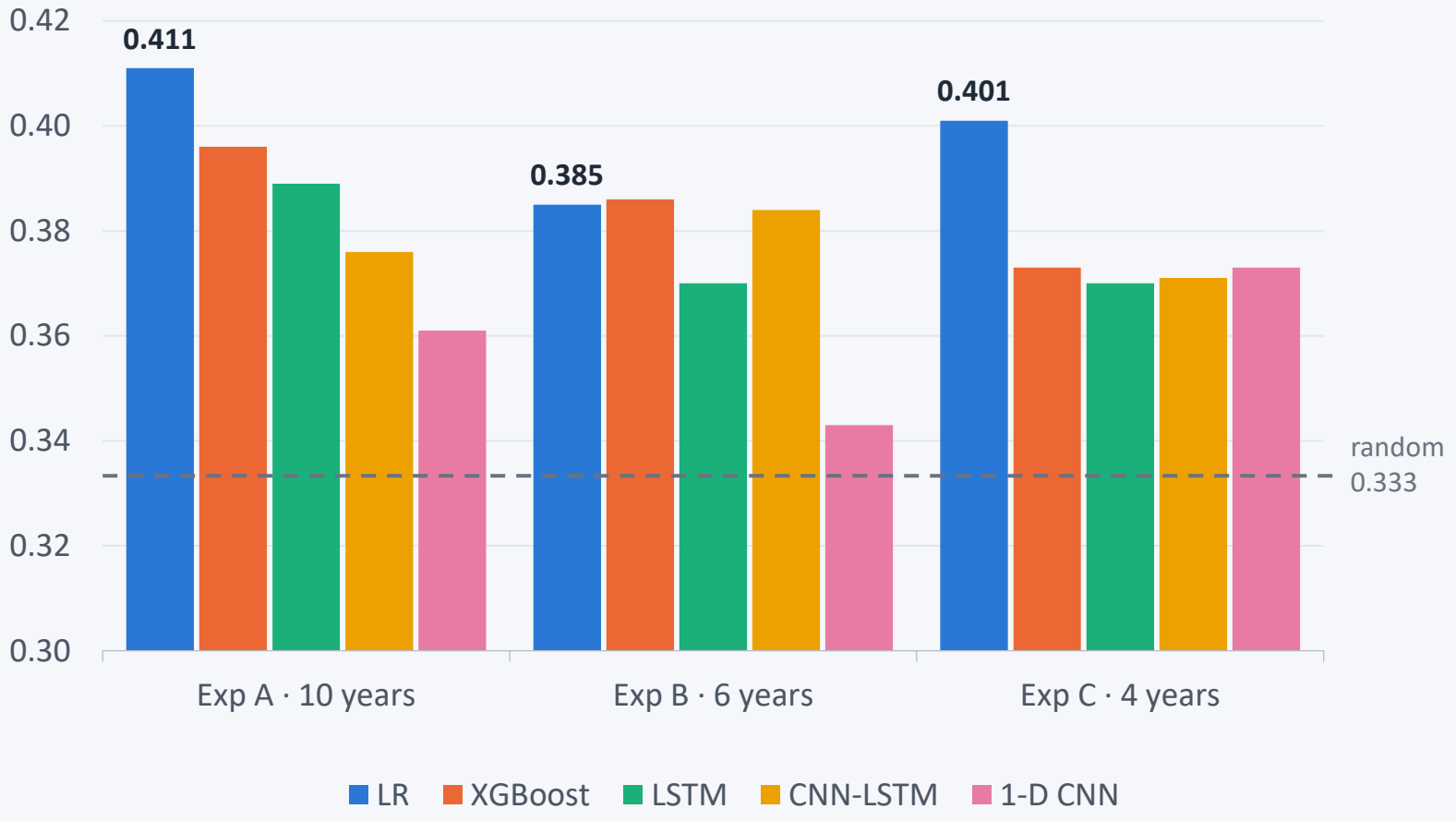
runs: 5 models × 3 splits × (1 global + 7 per stock)

Main metric: macro-F1

Plus: trading simulation (Sharpe, profit factor)

Result 1 · Logistic regression leads on average

Macro-F1 on the 2025 test year, global models (Table 3)



0.399

average macro-F1 of logistic regression (best)

Exp B · trading simulation

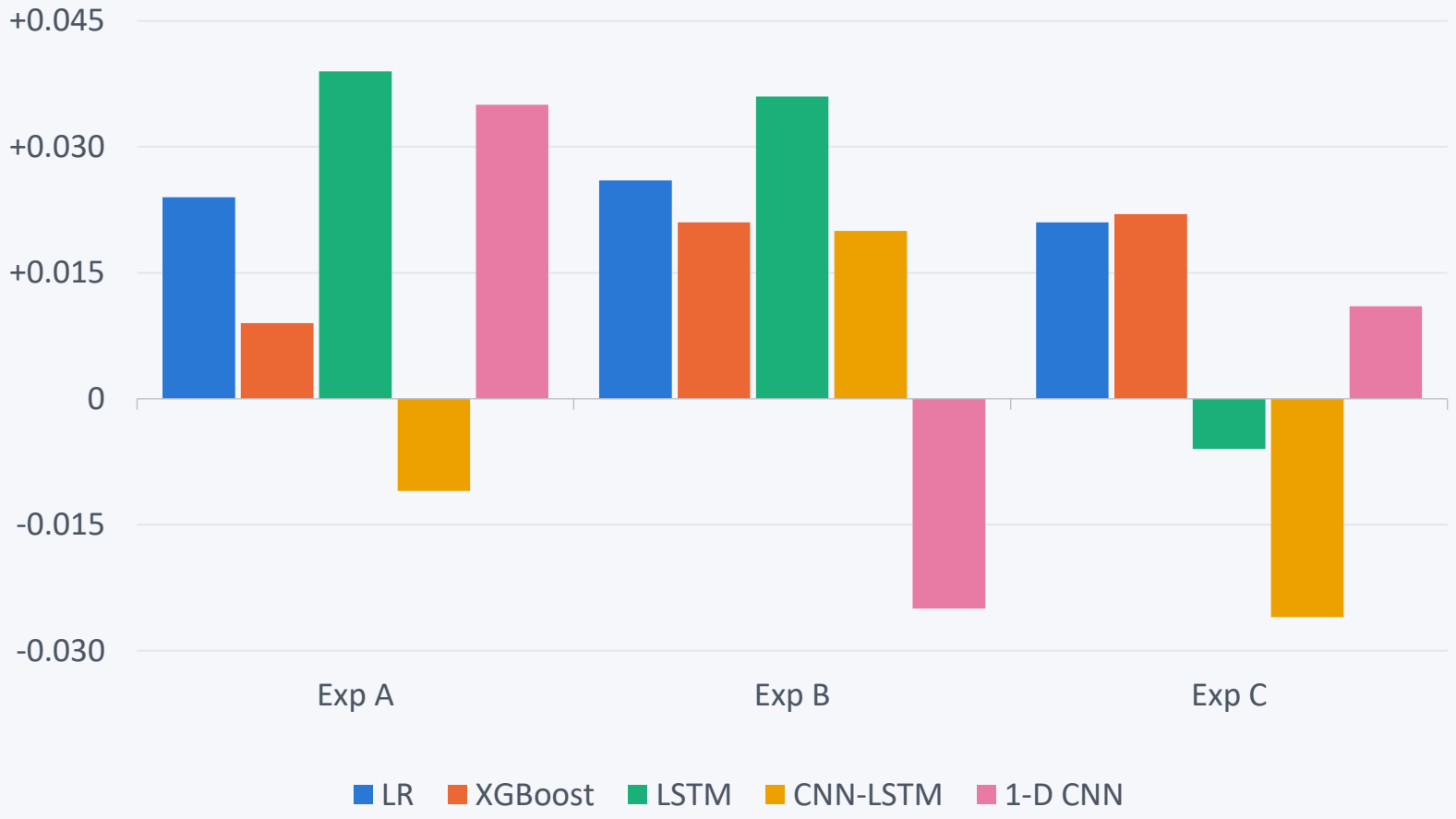
Sharpe **+0.76**

Profit factor **1.21**

both the highest of the five

Result 2 · Do not pre-filter the features

Δ macro-F1 = all 61 features − filtered features (Table 2)



11 / 15

cases where keeping all 61 features is better

Filters tested: Pearson + VIF, SHAP, Spearman

The model's own regularization selects better than an external filter.

Result 3 · The stock matters more than the model

Share of the variation of 105 per-stock Sharpe ratios (Section 7.3)

15.5 %

explained by the stock

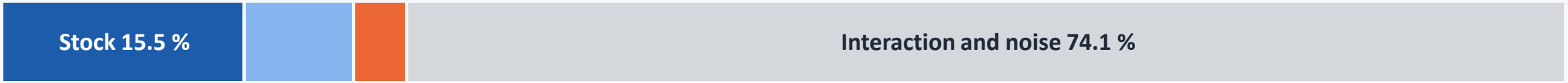
3.4 %

explained by the model

≈ 4.5×

more from the stock than from the model

Decomposition of the total sum of squares



Exp. 7.0 % · Model 3.4 %

AMZN does well with almost any model; GOOGL does badly with almost any model.

Was it fair? We rebuilt the protocol

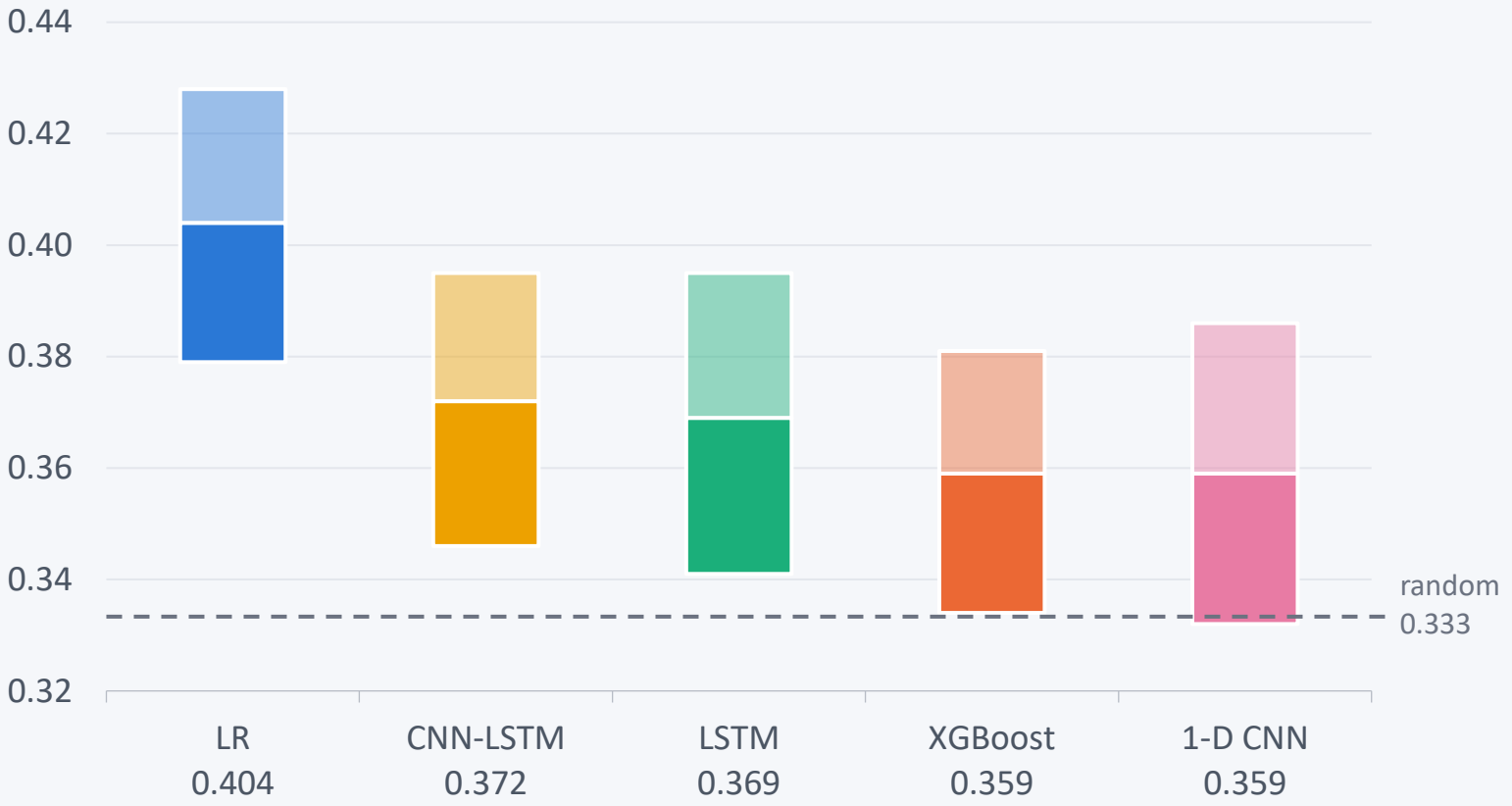
Reviewers' request → Section 8 of the paper



- ✓ **Same 1,736 test rows** for all five models (before: networks lost the first 20–60 days)
- ✓ **Same Optuna sampler** 150 trials (LR, XGBoost) · 80 trials, 17–20 hyper-parameters (networks)
- ✓ **No decision uses 2025** the test year is evaluated only once

Result 4 · The gap does not close

Exp B, corrected protocol: macro-F1 with 95 % block-bootstrap intervals (Table 6)



Dark part: lower bound → macro-F1 · light part: macro-F1 → upper bound

0.404

LR [0.379, 0.428] · significantly better than each of the other four

Networks: validation loss never clearly below ln 3 = 1.099

592 k → 10 k parameters: same result

Why does a linear model win here?

1

Small data

≈ 10,500 training rows vs. 10^5–10^6 network parameters

2

Noisy label

one-day direction is close to random

3

Rich features

indicators already summarize many days

Limitations

One test year (2025)

No transaction costs

Seven stocks, one sector

Same conclusion in the literature: tree and linear models match deep networks on tabular data (Grinsztajn et al., 2022); networks win with ~30,000 stocks and 60 years (Gu, Kelly and Xiu, 2020).

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Takeaways

1. A well-regularized logistic regression is a strong and cheap baseline.
2. Do not filter the features before training.
3. The stock matters more than the model.

Thank you · Questions?

Code and data: **github.com/Daivino/PAPER_MICAI**

Backup · Corrected protocol, Exp B (Table 6)

Model	Trials	Hyper-param.	Macro-F1	95 % CI	Δ vs. Sec. 7	Sharpe
LR	**150**	**5**	**0.404**	**[0.379, 0.428]**	**+0.019**	**+0.895**
CNN-LSTM	80	19	0.372	[0.346, 0.395]	−0.011	−0.074
LSTM	80	17	0.369	[0.341, 0.395]	−0.001	−0.345
XGBoost	150	9	0.359	[0.334, 0.381]	−0.027	+0.462
1-D CNN	80	20	0.359	[0.332, 0.386]	+0.016	−0.459

Paired differences of LR (all exclude zero): +0.045 XGBoost · +0.035 LSTM · +0.031 CNN-LSTM · +0.045 1-D CNN.

None of the six differences among the other four models is significant.

Biggest lever for the networks: choosing the epoch by validation F1 (+0.059 for the LSTM). Retraining them on train + validation makes them worse (−0.024 to −0.047).

Backup · Since submission (not in the paper)

Check	Result
Six test years, 2020–2025 (fixed configurations)	LR 0.360 · XGBoost 0.356 (difference not significant) · networks 0.326–0.338 (significantly lower)
Transaction costs, 10 basis points (Exp B, 2025)	Sharpe: LR +0.89 → +0.62 · XGBoost +0.46 → +0.01 (turnover 0.38 vs. 0.65)
Unseen data, Jan–Oct 2026	LR first in macro-F1 (0.346), not significant vs. XGBoost, LSTM, CNN-LSTM; 1-D CNN and LSTM had a higher Sharpe
Eight metrics on the same rows (2025)	LR best in all eight in A, B and C (F1, accuracy, balanced acc., kappa, AUC, log-loss, Sharpe 0 / 10 bp)

Same conclusion: classical regularized models match or beat the networks; LR vs. XGBoost is a tie over six years.

Backup · Benchmark tables (Section 7)

Global	A	B	C	Avg
LR	**0.411**	**0.385**	**0.401**	**0.399**
XGBoost	0.396	0.386	0.373	0.385
LSTM	0.389	0.370	0.370	0.376
CNN-LSTM	0.376	0.384	0.371	0.377
1-D CNN	0.361	0.343	0.373	0.359

Per stock	A	B	C	Avg
LR	**0.411**	**0.392**	**0.359**	**0.387**
XGBoost	0.376	0.370	0.352	0.366
CNN-LSTM	0.368	0.342	0.356	0.355
LSTM	0.314	0.353	0.326	0.331
1-D CNN	0.378	0.332	0.280	0.330

Exp B economics	F1	Win rate	Profit factor	Max DD	Sharpe
LR	**0.385**	**0.497**	**1.210**	**−0.470**	**+0.755**
XGBoost	0.386	0.500	1.083	−0.432	+0.345
CNN-LSTM	0.384	0.505	1.041	−0.403	+0.147
LSTM	0.370	0.491	1.120	−0.406	+0.433
1-D CNN	0.343	0.510	1.031	−0.434	+0.121

Backup · Hyper-parameter search (Section 8)

Model	Trials	Hyper-param.	Main search space	Selected
LR	150	5	penalty L1 / L2 / elastic-net, C ∈ [1e-4, 1e3], scaler	L2, C = 0.000165, robust scaler
XGBoost	150	9	trees, depth, learning rate, subsampling, L1/L2	485 trees, depth 9, lr 0.074
LSTM	80 (54 done)	17	layers, units, bidirectional, pooling, look-back, loss	2 × 118, attention, 10 days, focal loss
1-D CNN	80 (53 done)	20	blocks, filters, kernel, dilation, pooling, look-back	50 → 100 filters, kernel 7, 10 days
CNN-LSTM	80 (74 done)	19	conv filters, kernel, LSTM units, look-back	77 → 154 filters, LSTM 136, 20 days

Same TPE sampler (seed 42), same decision year (2024), objective macro-F1. One network training takes 7–12 s on a consumer GPU: compute was not the limit. Symmetric in procedure, not in dimension.